

Expt 1:

```
Enter text (single sentence or multiple sentences): Natural Language Processing is an exciting field of Artificial Intelligence.

Original Text:
Natural Language Processing is an exciting field of Artificial Intelligence.

Character Tokens:
['N', 'a', 't', 'u', 'r', 'a', 'l', ' ', ' ', 'L', 'a', 'n', 'g', 'u', 'a', 'g', 'e', ' ', ' ', 'P', 'r', 'o', 'c', 'e', 's', 's', 'i', 'n', 'g', ' ', ' ', 'i', 's', ' ', ' ', 'a', 'n', ' ', ' ', 'e', 'x', 'c', 'i', 't', 'i', 'n', 'g', ' ', ' ', 'f', 'i', 'e', 'l', 'd', ' ', ' ', 'o', 'f', ' ', ' ', 'A', 'r', 't', 'i', 'f', 'i', 'c', 'i', 'a', 'l', ' ', ' ', 'I', 'n', 't', 'e', 'l', 'l', 'i', 'g', 'e', 'n', 'c', 'e', ' ', '.']

Sentence Tokens:
['Natural Language Processing is an exciting field of Artificial Intelligence.']

Word Tokens:
['Natural', 'Language', 'Processing', 'is', 'an', 'exciting', 'field', 'of', 'Artificial', 'Intelligence', '.']

Stemmed Words:
['natur', 'languag', 'process', 'is', 'an', 'excit', 'field', 'of', 'artifici', 'intellig', '.']

Lemmatized Words:
['Natural', 'Language', 'Processing', 'is', 'an', 'exciting', 'field', 'of', 'Artificial', 'Intelligence', '.']

Comparison:
Stemming reduces words to root forms, which may not always be meaningful.
Lemmatization converts words into meaningful dictionary base forms.
```

Expt 2:

```
Enter a sentence: The quick brown fox jumps over the lazy dog.

Tokens:
['The', 'quick', 'brown', 'fox', 'jumps', 'over', 'the', 'lazy', 'dog', '.']

POS Tags:
The -> DT
quick -> JJ
brown -> NN
fox -> NN
jumps -> VBZ
over -> IN
the -> DT
lazy -> JJ
dog -> NN
. -> .

Tag Meanings:
NN -> Noun
VB -> Verb
JJ -> Adjective
RB -> Adverb
PRP -> Pronoun
DT -> Determiner

Total Words: 10
```

Expt 2 (HMM):

```
Enter a sentence: Artificial Intelligence improves healthcare and education.

POS Tags:
Artificial -> ``
Intelligence -> PRP
improves -> VBZ
healthcare -> VBN
and -> CC
education -> NN
. -> .
```

Expt 3 :

```
Enter number of headlines: 4
Enter headline: Apple launches new iPhone.
Enter headline: Samsung unveils latest Galaxy smartphone.
Enter headline: India wins the cricket world cup.
Enter headline: Virat Kohli scores a century.

Cosine Similarity Matrix:
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]

Headline Clusters:
Apple launches new iPhone. -> Cluster 0
Samsung unveils latest Galaxy smartphone. -> Cluster 0
India wins the cricket world cup. -> Cluster 0
Virat Kohli scores a century. -> Cluster 1

Enter first word: car
Enter second word: automobile
WordNet Similarity: 1.0
```

Expt 4:

```
Enter number of documents: 3
Enter document: Artificial Intelligence is transforming healthcare.
Enter document: Machine Learning improves prediction accuracy.
Enter document: Deep Learning is widely used in image recognition.

Enter search query: Machine Learning

TF-IDF Similarity Scores:
Document 1 : 0.0
Document 2 : 0.587
Document 3 : 0.172

LSA Similarity Scores:
Document 1 : -0.171
Document 2 : 0.995
Document 3 : 0.683

Most Relevant Document:
Machine Learning improves prediction accuracy.
```

Expt 5:

```
Enter legal text: The Supreme Court of India appointed Justice Chandrachud to hear the case in New Delhi.

Detected Named Entities:
Supreme -> ENTITY
Court -> ENTITY
India -> ENTITY
Justice -> ENTITY
Chandrachud -> ENTITY
New -> ENTITY
Delhi -> ENTITY

Enter actual number of entities: 4

Predicted Entities: 7
NER Accuracy: 57.14 %
```

Expt 6:

```
Enter biomedical sentence: Paracetamol treats fever and reduces pain.
Actual Relation (1 = Yes, 0 = No): 1

Tokens:
['paracetamol', 'treats', 'fever', 'and', 'reduces', 'pain', '.']

Predicted Relation: 1

Evaluation Results
-----
Precision : 1.0
Recall    : 1.0
F1-Score  : 1.0
```

Expt 7:

```
Enter a tweet: I love learning Natural Language Processing with Python.

Tokens:
['I', 'love', 'learning', 'natural', 'language', 'processing', 'with', 'python', '.']

===== N-GRAM MODEL =====

Unigrams:
[('I',), ('love',), ('learning',), ('natural',), ('language',), ('processing',), ('with',), ('python',), ('.',)]

Bigrams:
[('I', 'love'), ('love', 'learning'), ('learning', 'natural'), ('natural', 'language'), ('language', 'processing'), ('processing', 'with'), ('with', 'python'), ('python', '.')]

Trigrams:
[('I', 'love', 'learning'), ('love', 'learning', 'natural'), ('learning', 'natural', 'language'), ('natural', 'language', 'processing'), ('language', 'processing', 'with'), ('processing', 'with', 'python'), ('with', 'python', '.')]

Word Frequencies:
I : 1
love : 1
learning : 1
natural : 1
language : 1
processing : 1
with : 1
python : 1
. : 1

===== HMM MODEL =====
C:\Users\badri\OneDrive\Desktop\pacticals\venv\lib\site-packages\nltk\tag\hmm.py:333: RuntimeWarning: overflow encountered in cast
  X[i, j] = self._transitions[sj].logprob(self._states[j])
C:\Users\badri\OneDrive\Desktop\pacticals\venv\lib\site-packages\nltk\tag\hmm.py:335: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(sj, self._symbols[k])
C:\Users\badri\OneDrive\Desktop\pacticals\venv\lib\site-packages\nltk\tag\hmm.py:331: RuntimeWarning: overflow encountered in cast
  P[i] = self._priors.logprob(sj)
C:\Users\badri\OneDrive\Desktop\pacticals\venv\lib\site-packages\nltk\tag\hmm.py:363: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(sj, self._symbols[k])

HMM POS Tagging:
I -> NNP
love -> NNP
learning -> NNP
natural -> NNP
language -> NNP
processing -> NNP
with -> NNP
python -> NNP
. -> NNP

===== COMPARISON =====
N-Gram Model
- Learns word sequences.
- Predicts the next word based on previous words.
- Used for text generation and language modeling.

HMM Model
- Predicts Part-of-Speech (POS) tags.
- Uses transition and emission probabilities.
- Used for sequence labeling tasks.
```

Expt 8:

```
Enter number of reviews: 4
Enter review 1: The mobile phone has an excellent camera.
Enter review 2: Battery life is amazing and lasts all day.
Enter review 3: The laptop performance is very fast.
Enter review 4: The screen quality is excellent and bright.
```

Topics:

Topic 1

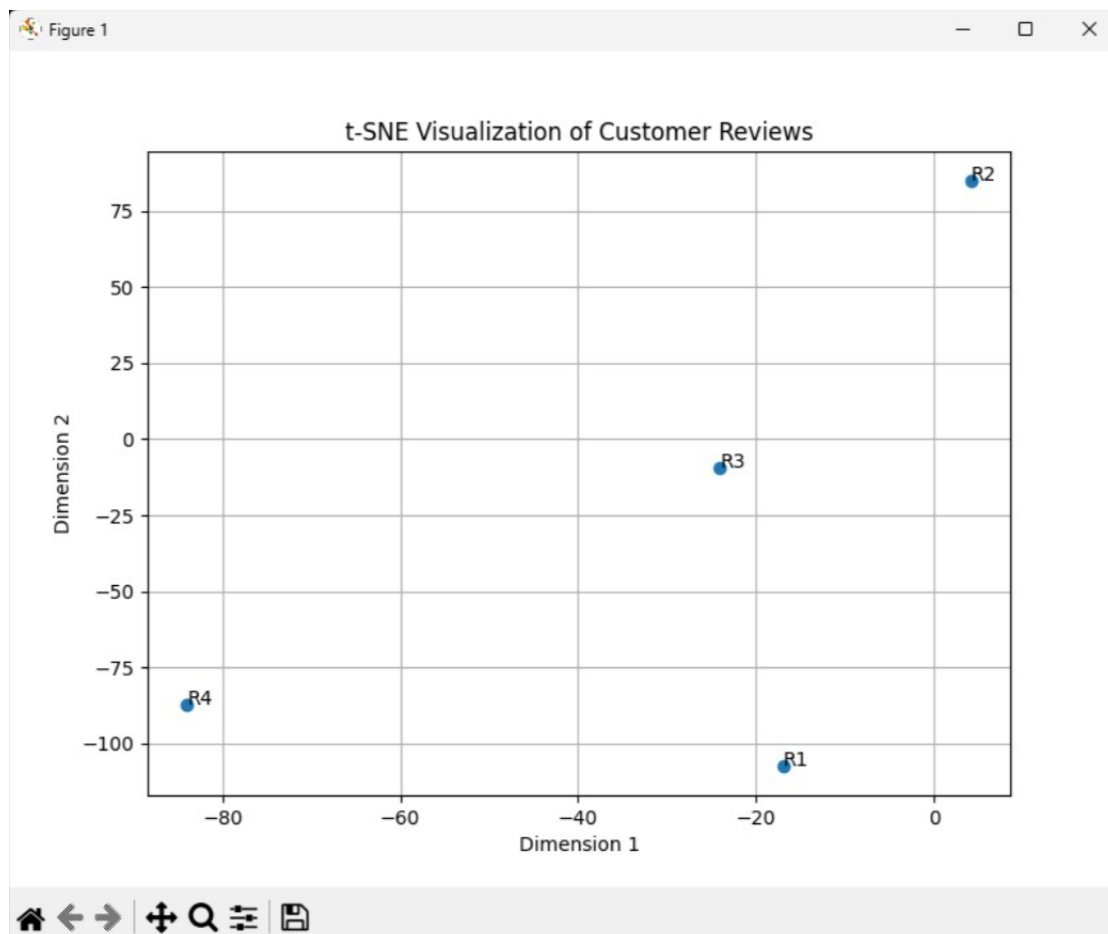
life
lasts
amazing
battery
day

Topic 2

screen
phone
camera
mobile
excellent

t-SNE Coordinates:

```
Review 1: [ -16.98687  -107.339226]
Review 2: [  4.179974  84.86165 ]
Review 3: [-24.075733  -9.262159]
Review 4: [-84.01048  -87.21845]
```



Expt 9

```
Enter number of documents: 3
Enter document: The contract was signed yesterday.
Enter category: contract
Enter document: The judgment was delivered by the court.
Enter category: judgment
Enter document: The agreement was accepted by both companies.
Enter category: agreement

Rule-Based Accuracy: 1.0
Maximum Entropy Accuracy: 1.0
```

Expt 10:

```
Enter number of posts: 4
Enter post: The new iPhone camera is amazing.
Enter post: Samsung Galaxy battery lasts longer.
Enter post: Football match was exciting today.
Enter post: Cricket world cup final was thrilling.
Enter number of clusters: 2
```

Cluster Results:

```
Post: The new iPhone camera is amazing.
Cluster: 0
```

```
Post: Samsung Galaxy battery lasts longer.
Cluster: 1
```

```
Post: Football match was exciting today.
Cluster: 0
```

```
Post: Cricket world cup final was thrilling.
Cluster: 0
```

Important Keywords:

```
Cluster 0
exciting
exciting today
football match
match exciting
amazing
```

```
Cluster 1
samsung
samsung galaxy
battery lasts
battery
galaxy
```

Marketing Insight:

```
Similar customer opinions are grouped together.
Clusters help identify product trends and issues.
```